

CFB GAGETOWN, NB, Canada

WMO: 717010

Lat: 45.8384N

Lon: 66.4331W

Elev: 166

StdP: 14.61

Time Zone: -4.00 (NAA)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-2.9	-20.7	1.8	-5.5	-15.3	2.4	-1.5	24.9	19.0	22.7	19.7	7.7	280	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	84.7	70.3	81.7	68.3	79.0	66.7	72.6	81.7	70.8	78.6	68.9	75.7	9.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	109.3	77.3	67.9	103.3	75.6	66.3	97.5	72.8	36.4	81.4	34.8	78.8	33.2	75.9	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.2	18.7	16.5	-14.7	90.9	5.3	2.3	-18.5	92.5	-21.7	93.8	-24.7	95.1	-28.5	96.8		
			-15.3	75.9	5.2	1.5	-19.0	77.0	-22.1	77.9	-25.0	78.7	-28.7	79.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	17.2	19.5	28.8	40.9	52.5	61.2	67.7	66.2	58.8	47.2	36.4	24.3	(6)
(7)		DBStd	19.30	11.67	10.67	9.49	7.25	6.73	6.32	4.99	5.16	7.28	7.66	8.79	10.20	(7)
(8)		HDD50	4237	1018	855	661	285	47	2	0	0	10	145	416	798	(8)
(9)		HDD65	8106	1482	1275	1124	724	393	148	29	47	210	552	859	1263	(9)
(10)		CDD50	1867	0	0	2	11	123	338	548	504	275	59	7	0	(10)
(11)		CDD65	260	0	0	0	0	5	33	111	85	25	1	0	0	(11)
(12)		CDH74	2063	0	0	3	7	100	328	853	590	174	8	0	0	(12)
(13)		CDH80	457	0	0	0	2	22	78	196	116	43	0	0	0	(13)
(14)	Wind	WSAvg	7.7	8.2	8.6	8.9	8.5	7.8	7.1	6.5	6.2	6.8	7.5	7.9	8.3	(14)
(15)	Precipitation	PrecAvg	45.3	4.1	3.4	3.8	3.4	3.8	3.3	3.5	3.6	3.7	3.8	4.4	4.6	(15)
(16)		PrecMax	59.8	8.8	6.5	7.6	6.1	8.3	7.8	6.9	9.4	10.7	7.2	8.1	8.1	(16)
(17)		PrecMin	33.3	0.8	1.1	1.1	0.8	0.8	1.0	1.1	0.7	0.8	0.8	1.7	1.9	(17)
(18)		PrecStd	6.4	1.7	1.4	1.4	1.3	1.9	1.6	1.4	2.0	1.8	1.7	1.5	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	49.4	60.1	70.6	82.6	87.0	88.6	87.4	85.6	74.6	63.3	54.7	(19)
(20)			MCWB	51.8	47.4	47.9	53.0	63.1	70.0	72.7	72.5	71.4	61.8	60.5	53.0	(20)
(21)		2%	DB	44.7	42.7	49.5	63.1	76.0	81.7	84.8	83.2	78.8	67.4	58.5	48.3	(21)
(22)			MCWB	43.3	39.4	42.1	50.3	60.2	66.7	70.9	70.0	67.6	59.5	55.0	46.9	(22)
(23)		5%	DB	38.5	38.6	45.6	58.0	70.6	77.7	81.8	80.1	74.3	63.5	54.4	42.9	(23)
(24)			MCWB	36.1	35.4	40.5	48.8	56.9	64.6	69.0	67.9	64.9	57.9	51.1	40.8	(24)
(25)		10%	DB	33.8	34.9	41.8	53.8	66.0	73.7	79.0	77.2	70.9	60.2	49.8	38.5	(25)
(26)			MCWB	32.0	32.2	36.4	46.0	54.3	62.2	67.4	66.4	63.1	55.1	46.8	36.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	48.0	51.8	58.9	67.2	73.0	75.1	74.3	72.8	65.3	60.9	53.7	(27)
(28)			MCDWB	53.0	48.9	55.9	64.6	79.8	83.8	84.6	83.9	82.9	70.5	62.7	54.4	(28)
(29)		2%	WB	43.6	40.5	44.9	54.5	62.6	69.5	72.8	72.1	69.5	62.0	56.3	47.0	(29)
(30)			MCDWB	44.5	42.2	47.2	59.0	71.6	77.7	82.1	80.2	75.6	65.8	58.1	48.2	(30)
(31)		5%	WB	36.4	35.8	40.5	50.4	59.3	66.7	71.0	70.2	67.0	59.1	51.8	41.5	(31)
(32)			MCDWB	38.1	38.3	44.7	56.3	66.9	74.3	79.2	77.1	72.2	63.2	53.7	42.7	(32)
(33)		10%	WB	32.1	32.6	37.0	46.6	56.2	64.2	69.0	68.3	64.7	55.6	46.6	36.5	(33)
(34)			MCDWB	33.6	34.6	41.2	52.8	63.9	71.5	76.2	74.3	69.1	59.2	49.2	37.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.3	17.3	16.9	17.9	20.6	19.9	19.7	19.6	20.4	17.2	13.9	14.0	(35)
(36)			MCDBR	21.2	19.5	21.2	26.8	29.4	26.2	23.5	23.4	22.8	21.8	19.0	18.4	(36)
(37)			MCWBR	21.2	17.1	16.0	17.0	16.3	14.2	12.1	12.3	13.5	15.3	16.9	18.5	(37)
(38)		5% WB	MCDWR	21.2	18.5	18.9	22.4	24.8	22.0	20.5	19.6	19.7	19.2	17.9	19.0	(38)
(39)			MCWBR	21.3	17.2	16.1	16.9	15.8	13.3	11.6	11.6	13.0	15.3	17.7	19.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.276	0.285	0.316	0.348	0.372	0.406	0.425	0.398	0.367	0.327	0.305	0.286	(40)	
(41)		taud	2.451	2.414	2.442	2.420	2.395	2.304	2.271	2.348	2.410	2.526	2.540	2.476	(41)	
(42)		Ebn at Noon	267	286	291	289	284	273	266	270	269	267	254	250	(42)	
(43)		Edh at Noon	22	27	31	34	36	40	41	36	32	24	20	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	475	748	1112	1377	1582	1711	1742	1543	1248	773	453	353	(44)	
(45)		RadStd	49	66	73	129	181	132	137	99	91	74	51	33	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	N/A

Nomenclature: See separate page